Data Visualization – Questions & Answers

Bar Chart · Histogram · Pie Chart · Scatter Plot (R / ggplot2)

Each question below is solved using **R** with **ggplot2**. The full R script is given, followed by the resulting chart and the requested written analysis, with all figures (means, medians, correlation, etc.) computed directly from the dataset.

```
install.packages(c("ggplot2", "dplyr"))
```

Question 1: Amounts – Bar Chart

Scenario: A supermarket chain wants to compare the monthly sales revenue generated by different product categories during June 2026, and identify the best and worst performing categories.

R Script

```
# ====================================================================
# Question 1: Bar Chart - Supermarket Category Sales (June 2026)
# ====================================================================
library(ggplot2)

categories <- c("Groceries", "Dairy", "Fruits", "Vegetables", "Bakery",
"Beverages", "Snacks", "Personal Care", "Household Items", "Frozen Foods")
sales <- c(120, 95, 80, 75, 68, 90, 110, 55, 70, 60)

df <- data.frame(Category = categories, Sales = sales)
df$Category <- factor(df$Category, levels = df$Category[order(df$Sales)]) # sort for plotting
df$Highlight <- ifelse(df$Sales == max(df$Sales), "Highest",
ifelse(df$Sales == min(df$Sales), "Lowest", "Other"))

p <- ggplot(df, aes(x = Category, y = Sales, fill = Highlight)) +
geom_col(width = 0.7) +
geom_text(aes(label = Sales), hjust = -0.15, size = 3.6) +
scale_fill_manual(values = c("Highest" = "#2E7D32", "Lowest" = "#C62828", "Other" = "#5B9BD5")) +
coord_flip() +
labs(title = "Monthly Sales Revenue by Product Category - June 2026",
x = NULL, y = "Sales (Rs. Lakhs)") +
theme_minimal(base_size = 13) +
theme(legend.position = "none", plot.title = element_text(face = "bold", hjust = 0.5)) +
ylim(0, max(df$Sales) * 1.15)

ggsave("q1_barchart_R.png", p, width = 9, height = 6, dpi = 150)

# ---- Answers (computed) ----
cat("Highest selling category:", df$Category[which.max(df$Sales)], "-", max(df$Sales), "Lakhs\n")
cat("Lowest selling category:", df$Category[which.min(df$Sales)], "-", min(df$Sales), "Lakhs\n")
print(df[order(-df$Sales), ])
```

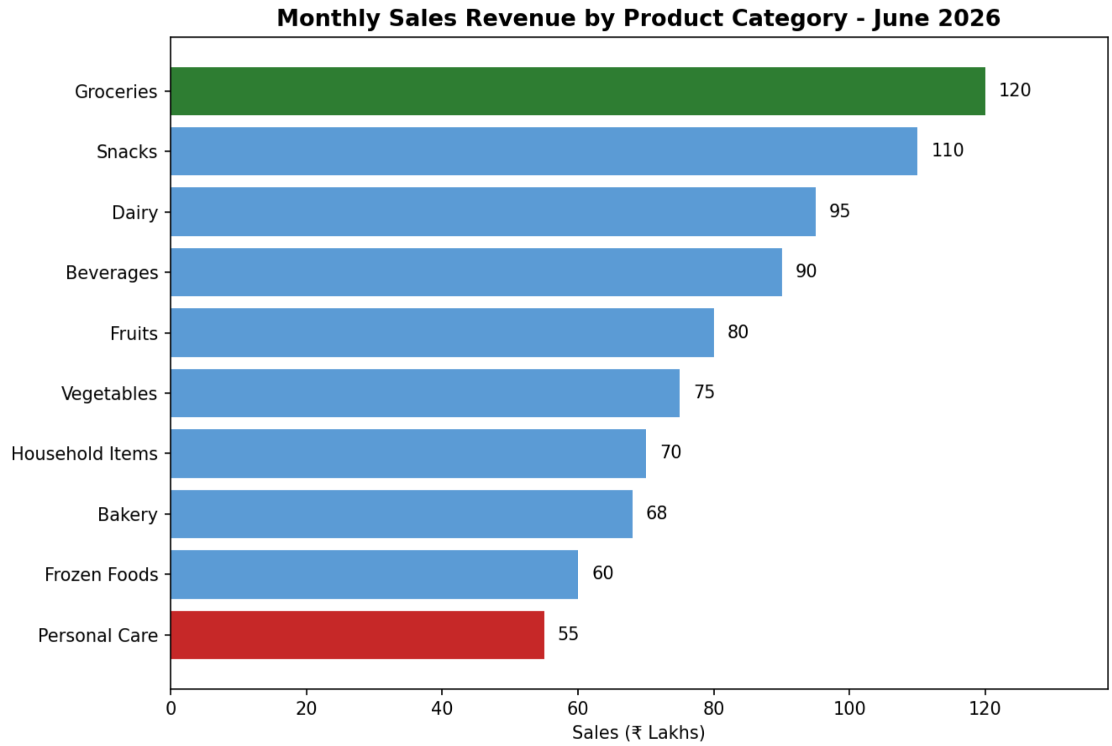

Figure 1.1 – Sales revenue by product category, June 2026 (highest in green, lowest in red).

Highest-selling category:

Groceries, with Rs. 120 Lakhs — the top performer by a clear margin.

Lowest-selling category:

Personal Care, with Rs. 55 Lakhs — the weakest performer.

Observation 1:

Everyday essentials dominate — **Groceries (120)** and **Snacks (110)** are the two highest earners, together contributing roughly 20% of total revenue across all ten categories, suggesting high-frequency, high-volume staples drive the bulk of supermarket sales.

Observation 2:

Discretionary/non-food categories lag behind — **Personal Care (55)**, **Frozen Foods (60)**, and **Bakery (68)** form the bottom tier, all under Rs. 70 Lakhs, roughly half of the top category's revenue, indicating an opportunity for targeted promotions or better shelf placement in these segments.

Question 2: Distribution – Histogram

Scenario: A university placement cell collected the annual salary packages (in LPA) offered to students during campus recruitment, and wants to study the distribution of these packages.

R Script

```
# =====================================================================
# Question 2: Histogram - Salary Package Distribution (LPA)
# =====================================================================
library(ggplot2)

salary <- c(3.2, 4.5, 5.0, 6.8, 7.2, 8.5, 4.8, 5.5, 6.0, 7.8, 9.2, 10.5)
df <- data.frame(Salary = salary)

p <- ggplot(df, aes(x = Salary)) +
geom_histogram(binwidth = 1.5, boundary = 3, fill = "#4C72B0", color = "white") +
geom_vline(aes(xintercept = mean(Salary)), color = "#C62828", linetype = "dashed", linewidth = 1) +
labs(title = "Distribution of Salary Packages (LPA)",
x = "Salary Package (LPA)", y = "Frequency (No. of Students)") +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", hjust = 0.5))

ggsave("q2_histogram_R.png", p, width = 8.5, height = 6, dpi = 150)

# ---- Answers (computed) ----
cat("Mean:", mean(salary), " Median:", median(salary), " SD:", sd(salary), "\n")
cat("Range:", range(salary), "\n")
q <- quantile(salary, c(0.25, 0.75))
iqr <- IQR(salary)
cat("Q1:", q[1], " Q3:", q[2], " IQR:", iqr, "\n")
lower_fence <- q[1] - 1.5 * iqr
upper_fence <- q[2] + 1.5 * iqr
cat("Outlier fences:", lower_fence, "-", upper_fence, "\n")
outliers <- salary[salary < lower_fence | salary > upper_fence]
cat("Outliers:", outliers, "\n")
```

Figure 2.1 – Distribution of salary packages (LPA), dashed line = mean.

Spread:

Packages range from **3.2 to 10.5 LPA** (range = 7.3), with mean ≈ **6.58 LPA**, median = **6.4 LPA**, and standard deviation ≈ **2.15 LPA**. The closeness of mean and median indicates a fairly balanced, only mildly right-skewed spread.

Frequency:

The histogram (bin width 1.5 LPA) shows the largest concentration of students in the **4.5–7.5 LPA** range, covering the majority of offers, with progressively fewer students in the lower (below 4.5) and higher (above 9) brackets — a roughly bell-shaped, unimodal pattern.

Outliers:

Using the IQR method (Q1 ≈4.95, Q3 ≈7.975, IQR ≈3.03), the acceptable range is ≈**0.41 to 12.51 LPA**. Since every value (3.2 to 10.5) falls inside this range, **there are no statistical outliers** — the highest offer (10.5 LPA) is a strong package but not an anomaly relative to the group.

Question 3: Proportions – Pie Chart

Scenario: A software company allocated its annual digital marketing budget across different advertising channels, and wants to visualize how the budget is distributed.

R Script

```
# ======================================================================
# Question 3: Pie Chart - Digital Marketing Budget Allocation
# ======================================================================
library(ggplot2)
library(dplyr)

channels <- c("Social Media", "Television", "Print Media", "Radio",
"Email Marketing", "Influencer Marketing", "SEO", "Events")
budget <- c(28, 22, 10, 8, 12, 9, 6, 5)

df <- data.frame(Channel = channels, Budget = budget)
df <- df %>%
arrange(desc(Budget)) %>%
mutate(Channel = factor(Channel, levels = Channel),
Label = paste0(Channel, " (", Budget, "%)"),
ypos = cumsum(Budget) - 0.5 * Budget)

p <- ggplot(df, aes(x = "", y = Budget, fill = Channel)) +
geom_bar(stat = "identity", width = 1, color = "white") +
coord_polar(theta = "y") +
geom_text(aes(label = paste0(Budget, "%")), position = position_stack(vjust = 0.5), size = 3.5) +
scale_fill_brewer(palette = "Set3") +
labs(title = "Digital Marketing Budget Allocation by Channel", fill = "Channel") +
theme_void(base_size = 13) +
theme(plot.title = element_text(face = "bold", hjust = 0.5))

ggsave("q3_piechart_R.png", p, width = 8.5, height = 6.5, dpi = 150)

# ---- Answers (computed) ----
print(df[order(-df$Budget), c("Channel", "Budget")])
cat("Highest investment channel:", as.character(df$Channel[which.max(df$Budget)]),
"-", max(df$Budget), "%\n")
```

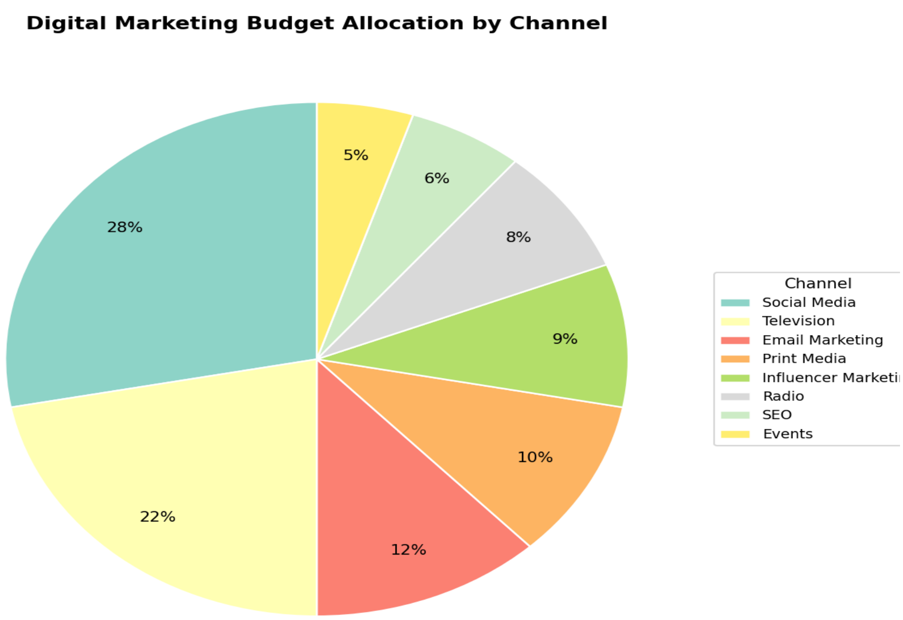

Figure 3.1 – Digital marketing budget allocation by channel.

Channel with highest investment:

Social Media, receiving **28%** of the total budget — the single largest slice.

Overall distribution:

Spending is concentrated in digital-first, high-reach channels: **Social Media (28%)** and **Television (22%)** together account for **50%** of the entire budget. A second tier — **Email Marketing (12%)**, **Print Media (10%)**, and **Influencer Marketing (9%)** — makes up another 31%. The remaining channels, **Radio (8%)**, **SEO (6%)**, and **Events (5%)**, receive the smallest shares, together just 19%. This reflects a strategy weighted toward broad-reach and social/digital engagement over niche or long-term organic channels like SEO and offline events.

Question 4: X–Y Relationship – Scatter Plot

Scenario: An e-commerce company wants to determine whether advertising expenditure influences monthly sales revenue, based on eight months of data (January–August).

R Script

```
# ======================================================================
# Question 4: Scatter Plot - Advertising Cost vs Sales Revenue
# ======================================================================
library(ggplot2)

month <- c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug")
ad_cost <- c(5, 8, 10, 12, 15, 18, 20, 22)
sales_revenue <- c(42, 55, 63, 70, 82, 91, 98, 110)

df <- data.frame(Month = month, AdCost = ad_cost, Sales = sales_revenue)

p <- ggplot(df, aes(x = AdCost, y = Sales)) +
geom_point(size = 3.5, color = "#2E5395") +
geom_smooth(method = "lm", se = FALSE, color = "#C62828", linetype = "dashed") +
geom_text(aes(label = Month), vjust = -1, size = 3.3) +
labs(title = "Advertising Cost vs Sales Revenue",
x = "Advertising Cost (Rs. Lakhs)", y = "Sales Revenue (Rs. Lakhs)") +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", hjust = 0.5))

ggsave("q4_scatterplot_R.png", p, width = 8.5, height = 6, dpi = 150)

# ---- Answers (computed) ----
correlation <- cor(ad_cost, sales_revenue)
model <- lm(Sales ~ AdCost, data = df)
cat("Correlation coefficient (r):", round(correlation, 4), "\n")
print(summary(model))
```

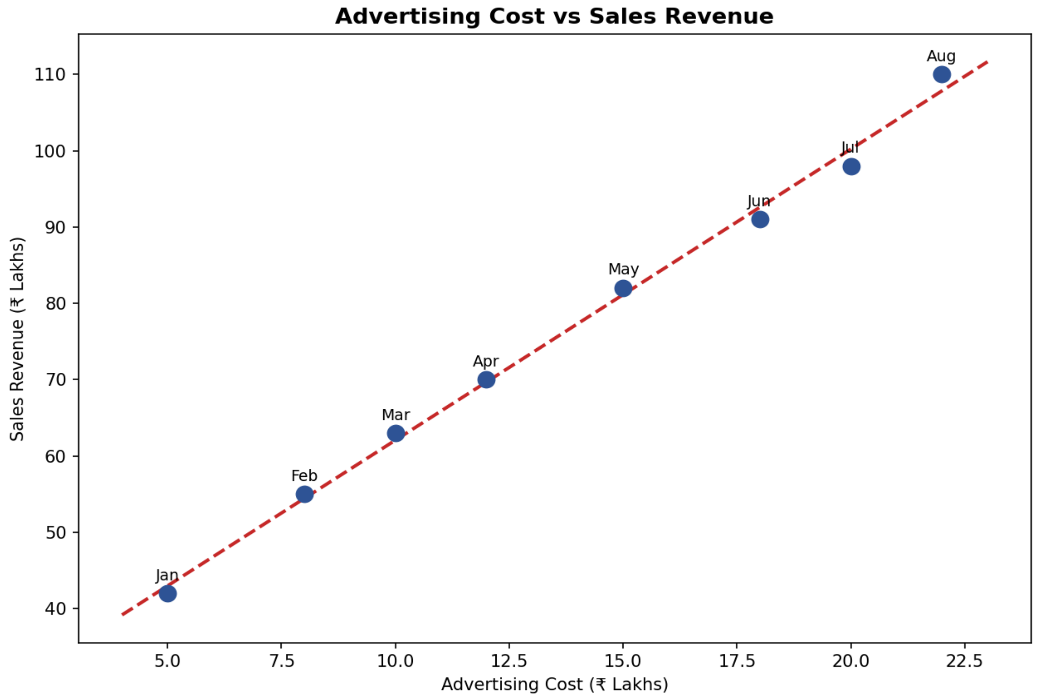

Figure 4.1 – Advertising cost vs sales revenue with fitted linear trend line.

Relationship:

The scatter plot shows a clear, consistent **positive linear relationship** — as advertising spend rises from Rs. 5 to Rs. 22 Lakhs, sales revenue rises steadily from Rs. 42 to Rs. 110 Lakhs, with points lying almost exactly along the fitted trend line.

Correlation coefficient:

cor(AdCost, Sales) = **r** ≈ **0.998** ($R^2 \approx 0.996$) — an extremely strong positive correlation, indicating advertising expenditure explains almost all of the month-to-month variation in sales revenue in this dataset.

Trend equation:

The fitted line is approximately **Sales ≈ 3.81 × AdCost + 23.94**, meaning each additional Rs. 1 Lakh spent on advertising is associated with roughly **Rs. 3.81 Lakhs** more in sales revenue.

Interpretation:

The near-perfect linear fit strongly suggests advertising expenditure is a major driver of sales in this period. However, with only 8 monthly data points and no controls for other factors (seasonality, pricing, promotions), this is a strong **association**, not proof of causation — further testing (e.g., controlled experiments or a longer time series) would be needed to confirm a causal effect.